

Global Map of Folklore

With the development of technology, folkloristics materials (e.g. tales, myths, gossip, rumours, urban legends), either as new emergence or variations of old (or also recent) stories, have been relentlessly populated. Given the rapid increase of narratives, it has been suggested by the scholars (Abello et al., 2012) to adopt computational methods as a way to study and engage with the multiplicity and complexity of narratives circulating in contemporary cultures. This project pursues and is hence an attempt to materialise that spirit. I developed a web that collects urban legends and displays the locations where these stories probably originated or was distributed to. The web includes features for performing NLP analysis to help users analyse or develop insights on the narrative materials.

Introduction

The idea for this project sparked from my interest in folklore studies, a scholar field that investigates expressive forms of traditional or vernacular cultures “to understand how traditional expressions create meaning for the people who create and receive them” (Abello et al., 2012, p. 60). For this project, my attention is particularly paid to urban legends “a secondhand story of doubtful origins but nonetheless told as true” (Wiecko, 2014). I have noticed that many cultures share similar stories that have particular recurring elements. For instance, the “vanishing hitchhiker” — a narrative involving a driver picking up a mysterious hitchhiker that eventually turned out to be a ghost or supernatural entity — is told not only in the US but also common in Japan and Thailand. This similarity in the narrative despite physical distance ignites the idea of this project. The aim is to visualise the locations where folklore probably originated from or spreads in order to demonstrate universality of narratives across various cultures, helping facilitate researchers or general users analyse and develop insights on the collected narratives.

Method

Folkloristic analysis. First, I collected urban legends that mention locations of probable origins or of diffusion. Then, I identified motifs (recurring elements in stories) and themes in those stories using indices proposed by previous studies — Thompson's (1955-58) *Motif-Index of Folk-Literature* and Brunvand's (1981) index of urban and contemporary legends — as well as my own interpretations. The identified elements were used as tags to characterise the gathered narratives.

Web development. The web was developed in VS Code. For the features that this web attempts to attain — searching and filtering, displaying an interactive map, and performing NLP tasks using JavaScript — were beyond my current understanding of web programming, ChatGPT by OpenAI was also used in this process of web development.

Development Process

To begin the project, I browsed and explored existing websites for inspiration, both of content and web design. The draft idea and prototype were presented to my instructor to have some feedback. Based on the outline and given advice, the web code was developed by providing prompts to ChatGPT to include complex features that are beyond my current knowledge of web building such as search bar and filtering tags, interactive map, and functions for performing natural language processing. The web content (e.g. collected urban legends) was created according to the method mentioned above and stored as a JSON file. Then, together, the web code and content were deployed as a Github page.

Program Design

On the top lies the web header, specifying its title and subtitle. On the left side of the screen posit the search bar, tags/motifs which can be clicked and selected to filter stories, and story cards. On each card, the story name and its relevant tags are shown. At the center of the screen rest the global map with pins corresponding to the stories' locations. Below the map is a panel displaying the full story when a story card is selected. On the right side of the screen are the NLP analysis features. One of these includes a panel displaying a graph of stories that are semantically related to the story selected. Below this panel is a smaller panel showing keywords (i.e. most frequent words found in a story) as well as named entities (if detected).

Evaluation

Evaluating the web performance myself, I contend that the web can function on bare minimum. For the search bar and filtering tags, when a tag was selected, story cards with that tag were displayed, successfully filtering stories. However, when inputting a search term, the searching function occasionally returned outputs irrelevant to the term. For instance, when inputting the term “cow”, the story card *Heavy School Books* was returned although the story does not involve the animal.

The semantic connection feature did provide impressive results. When a story was selected, the stories shown to be semantically connected to that story did really relate. Yet, an issue for this feature lies in its interface. The displayed texts and labels were relatively small and sometimes overlapped each other, making the graph hard to read.

The tokenisation function performed well. It could extract and display the most frequent terms of a story. Yet, a main pain point was at the NER function. The feature mostly could not recognise a location’s name. For instance, it could not extract the name “Beijing” although the term was explicitly mentioned in the story *Bus 375*. It also could not recognize a person’s name. For example, it could not identify the name “Marissa”, which is the name of a person explicitly mentioned in the story *Hormones in Cow’s Milk*.

Additionally, due to limited time, no user testing was carried out. Hence, there are currently no users’ perspectives, and this evaluation is subjective. I acknowledge this as a limitation of the project.

Limitations and Suggestions

The limitations of this project lie not only in the technical aspect of the web but also in the analysis and content. Most legends collected in this web are predominantly American since the available and searchable data with mentioning of locations were mostly published and found in American media. Hence, the project, as is now, cannot fulfill the objective to illustrate commonality of narratives across different cultures. It is at the same time advisable that the prospective future should more gather stories of sources in various cultures.

Moreover, the identification of motifs or themes as part of the analysis process relied solely on my subjective point of view. As I am the only person doing the analysis of the legends, my interpretations could be wrong or miss some details. This suggests cooperation from scholars or experts of folkloristics.

For the web technical aspect, I would suggest improvements in features and interface, especially the NER function. A possible feature to include is the time dimension of stories.

References

- Abello, J., Broadwell, P., & Tangherlini, T. R. (2012). Computational folkloristics. *Communications of the ACM*, 55(7), 60–70. <https://doi.org/10.1145/2209249.2209267>
- Brunvand, J. H. (1981). *The Vanishing Hitchhiker: American urban legends and their meanings*. W. W. Norton & Company.
- Thompson, S. (1955-58). *Motif-Index of Folk-Literature* (Vs. 1–6). Indiana University Press.
- Wiecko, F. (2014). Urban Legends. *Encyclopedia of Social Deviance*. <https://doi.org/10.4135/9781483340470.n306>

References for Web Design

- [Database of Danish Folklore](#)
- [The Folklore Database](#)
- [Urban Legends of The United States](#)